

BALM: Boundary-Aligned Logit Margin Learning for Zero-Shot Industrial Anomaly Detection and Segmentation

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Abstract

Zero-shot industrial anomaly detection and segmentation aim to identify and localize unseen defects without target-domain training data, enabling rapid deployment on new production lines. Recent vision foundation model based methods achieve strong generalization, yet they still produce continuous anomaly maps whose final segmentation quality heavily depends on post-hoc binarization. This mismatch between training objectives and inference-time decision boundaries remains largely overlooked. To address it, we propose Boundary-Aligned Logit Margin Learning (BALM), which introduces threshold-calibrated learning into zero-shot industrial anomaly detection and segmentation. BALM shifts the default logit decision boundary from the conventional probability threshold of 0.5 to the target operating threshold, and further enforces a shifted-logit margin to better separate normal and anomalous pixels around the calibrated boundary. We further refine the resulting binary masks with the Segment Anything Model (SAM) to obtain sharper anomaly boundaries. Experiments on MVTec AD 2 show that BALM consistently improves zero-shot anomaly detection and segmentation under challenging industrial conditions, and that SAM-based refinement further boosts defect localization quality. Code is available at [link].

1. Introduction

Visual anomaly detection (AD) is a fundamental task in industrial inspection, aiming to identify abnormal visual pat-

terns and localize defective regions [1, 8, 18, 21]. Most existing AD methods assume that sufficient normal images from the target domain are available for training [5, 15, 17]. However, this assumption is often impractical when new production lines are deployed, as in-domain samples may be unavailable or costly to collect. Zero-shot AD is therefore of practical importance, as it aims to detect and localize defects without requiring target-domain training data.

Recent zero-shot methods commonly leverage vision-language models (VLMs), such as CLIP [16], by aligning visual features with textual prompts or learned tokens that represent generic concepts of normality and abnormality [3, 4, 10, 19, 20]. Although image-text pretraining provides useful semantic priors for category-level generalization, textual descriptions are often too coarse to capture fine-grained industrial defects, such as tiny scratches, contamination, or changes in transparency. In contrast, vision foundation models (VFM) provide dense visual representations that are better suited to capturing local appearance deviations. However, prior VFM-based adaptations often train only a shallow prediction head, leaving the internal visual representations largely unchanged, which limits their ability to distinguish subtle anomalies from normal appearance variations across unseen objects [7, 9].

AnomalyVFM addresses this limitation by adapting VFMs for zero-shot anomaly detection through synthetic anomaly training and parameter-efficient feature adaptation [6]. Building on this framework, our work focuses on a critical issue for deployment that remains underexplored: how to convert continuous anomaly maps into reliable binary masks without target-domain validation data. In standard AD settings, the binarization boundary can be tuned using validation samples or target-domain statistics. Under

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the zero-shot protocol, however, such calibration is unavailable, making the decision boundary a key determinant for pixel-level segmentation performance.

1.1. Challenge Description

This challenge addresses zero-shot industrial anomaly detection and segmentation, where anomalous regions must be localized in unseen test images without access to target-domain training or validation data. The difficulty mainly lies in two aspects:

- **Complex industrial conditions.** MVTec AD 2 covers eight industrial products and presents diverse real-world challenges, including occlusion, material variation, reflective and translucent surfaces, extreme illumination, grayscale imagery, high-resolution inputs, and non-uniform aspect ratios. These factors substantially increase the difficulty of robust anomaly localization across categories.
- **Fixed-threshold segmentation under zero-shot constraints.** Practical inspection systems require binary defect masks rather than continuous anomaly scores. However, in the zero-shot setting, the binarization threshold cannot be tuned using target-domain statistics, validation samples, or category-specific calibration. A central challenge is therefore to ensure that a predefined decision boundary remains effective across unseen categories and imaging conditions.

2. Methodology

In this section, we describe the BALM method for zero-shot visual anomaly segmentation. We first present the overall model design and explain how our method builds upon AnomalyVFM. We then introduce the threshold-calibrated segmentation objective, followed by the training strategy, post-processing procedure, and evaluation protocol.

2.1. Model Design

The proposed model retains the general structure of AnomalyVFM while modifying the segmentation objective to better match the fixed threshold used during inference. This design allows the model to benefit from anomaly-sensitive VFM adaptation while explicitly improving the reliability of binary mask generation under the zero-shot protocol.

2.1.1. Approach

We build our solution on AnomalyVFM, which adapts vision foundation models (VFMs) for zero-shot industrial anomaly detection through synthetic anomaly training and parameter-efficient feature adaptation. We choose this approach because industrial defects are mainly reflected by subtle local appearance changes, such as texture inconsistencies, structural disruptions, and weak boundary deviations, which require strong visual representations and dense

localization ability. Compared with vision–language methods that rely more on coarse semantic priors, VFM-based modeling is better suited to this challenge.

AnomalyVFM provides a strong starting point by exposing the model to diverse synthetic defect patterns during training and adapting the backbone with lightweight modules for anomaly-sensitive feature learning. This design offers good zero-shot generalization without requiring target-domain data.

2.1.2. Architecture

Our architecture follows AnomalyVFM, while our main modification is a threshold-calibrated segmentation objective. The default probability threshold of 0.5 is often too high for zero-shot anomaly segmentation, since anomalous regions are usually low-contrast and highly imbalanced with normal pixels, causing many true anomalous pixels to receive relatively low confidence scores. Moreover, under the zero-shot setting, the decision boundary cannot be tuned with target-domain validation data. Therefore, during training, we shift the default decision boundary to a lower fixed threshold and explicitly align the learning objective with the operating threshold used at inference.

Training-time threshold calibration. Let p_0 denote the default operating threshold used during training, which is set to $p_0 = 0.05$ in our implementation. We convert it to the corresponding logit boundary

$$b = \log \frac{p_0}{1 - p_0}. \quad (1)$$

For $p_0 = 0.05$, this gives $b \approx -2.9444$. Given the predicted anomaly logits M , we define the shifted logits as

$$\widetilde{M} = M - b, \quad (2)$$

so that the zero point in the shifted logit space corresponds to the calibrated operating boundary rather than the default probability boundary of 0.5.

Pixel-level supervision is then applied to $\widetilde{M}$ using a boundary-aware confidence-weighted focal and L1 objective. Specifically, the base pixel loss is

$$\mathcal{L}_{\text{base}} = 5 \mathcal{L}_{\text{focal}}(\widetilde{M}, Y) + \left\| \text{clip}\left(\sigma(\widetilde{M}), 0.1, 0.9\right) - Y \right\|_1, \quad (3)$$

where Y is the ground-truth anomaly mask and $\sigma(\cdot)$ denotes the sigmoid function. Following the confidence-weighted design of AnomalyVFM, the decoder predicts a confidence map c , which is transformed as

$$C = 1 + \exp(c). \quad (4)$$

The segmentation loss is computed element-wise as

$$\mathcal{L}_{\text{seg}} = \text{mean}(C \odot \mathcal{L}_{\text{base}} - \alpha \log C), \quad (5)$$

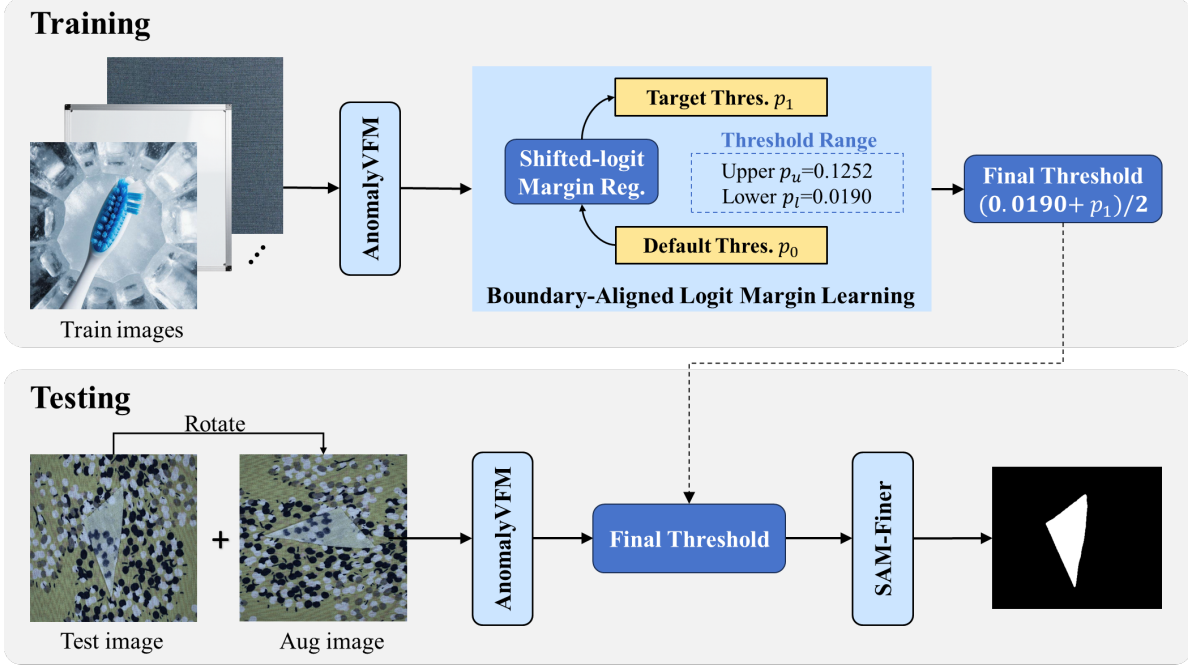


Figure 1. Overview of the boundary-calibrated AnomalyVFM framework for zero-shot visual anomaly segmentation.

where $\alpha = 0.1$ and $\odot$ denotes element-wise multiplication.

To further enlarge the separation between normal and anomalous pixels around the calibrated boundary, we introduce a shifted-logit margin regularization term:

$$\mathcal{L}_{\text{margin}} = Y \cdot \text{softplus}(m_{\text{pos}} - \widetilde{M}) + (1 - Y) \cdot \text{softplus}(\widetilde{M} + m_{\text{neg}}). \quad (6)$$

This term encourages anomalous pixels to lie above the calibrated boundary and normal pixels to lie below it. The overall training objective is

$$\mathcal{L} = \mathcal{L}_{\text{img}} + \mathcal{L}_{\text{seg}} + \gamma \mathcal{L}_{\text{margin}}, \quad (7)$$

where $\mathcal{L}_{\text{img}}$ is the image-level focal loss and γ controls the contribution of the margin term.

Inference-time threshold adjustment. Training is performed with a fixed operating threshold p_0 , while inference adopts a more recall-oriented threshold to reduce missed defects. Specifically, we set the default threshold to $p_0 = 0.05$ and define a lower bound $p_l = 0.0190$. The final inference threshold is chosen as the midpoint between them:

$$p_{\text{final}} = \frac{p_0 + p_l}{2}. \quad (8)$$

This shifts the operating point slightly below the training threshold, making the model more sensitive to weak or small anomalous regions and thereby improving defect recall. With this design, training remains aligned with a fixed calibrated boundary, while inference uses a slightly more permissive threshold for practical defect localization.

2.1.3. Training

Our method follows the zero-shot protocol and uses only auxiliary synthetic anomaly data for training, no MVTec AD 2 images are used for optimization or validation.

All input images are resized to 672×672 . We do not split images into local patches, since patch-wise processing was found to introduce noisi predictions in the zero-shot setting. We use TIPSv2-so400m14 [2] as the backbone and apply DoRA-based parameter-efficient adaptation [13] with rank 64. The model is trained for 2,500 iterations using AdamW [14] with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-2} . The batch size is 16, implemented with gradient accumulation over 4 steps. The random seed is fixed to 12. During inference, a fixed pixel-level boudary of $p_{\text{final}} = 0.0345$ is used to obtain the final binary anomaly masks.

2.2. Post-processing

We directly adopt the SAM-Finer strategy from RoBiS [12] as our post-processing module. Specifically, we use the coarse binary anomaly mask as a localization prior and leverage SAM [11] to refine object-aware boundaries. In our implementation, we use the `sam_vit_h` model by default and extract the bounding rectangles of up to three connected components from the binary mask as box prompts for SAM. This refinement procedure is applied to all categories in our experiments.

2.3. Dataset & Evaluation

We conduct experiments on the MVTec AD 2 benchmark under the official industrial zero-shot evaluation protocol. The benchmark images are used exclusively for evaluation, with no target-domain training or validation samples involved in model optimization or threshold selection. During evaluation, all images are resized to 672×672 . We do not apply patch-wise cropping, as preserving global image context is important for stable zero-shot anomaly localization. Instead, we adopt a simple test-time augmentation strategy by rotating each image by 0° and 90° , performing inference on both views, and averaging the resulting anomaly predictions. To further improve the spatial consistency of the predicted anomaly maps, we apply average-pooling smoothing with a mean kernel size of 5.

We strictly follow the official evaluation protocol, where the pixel-level F1 score (SegF1) is used as the primary metric. Our model directly predicts the final binary anomaly masks, which matches the requirement of the zero-shot track: to localize anomalies and produce binary segmentation results without using any target-domain information.

3. Results

We evaluate our method on the original MVTec AD 2 benchmark under the official zero-shot setting. Since the task requires binary anomaly segmentation, we report the pixel-level F1 score, denoted as SegF1, as the primary performance metric. Then, we compare it with conventional anomaly detection baselines to highlight the effectiveness of our method.

3.1. Performance Metrics

We evaluate our method on the MVTec AD 2 benchmark under the official zero-shot protocol, using the pixel-level F1 score (SegF1) as the primary metric. Our method achieves a SegF1 of **42.64 %** on $\text{TEST}_{\text{priv}}$ and **41.45 %** on $\text{TEST}_{\text{priv,mix}}$. The small performance gap between the two test sets indicates that the proposed method remains stable under mixed and more challenging evaluation conditions, suggesting good robustness.

A particularly notable result is obtained on the *fabric* category, where our method reaches **94.10 %** on $\text{TEST}_{\text{priv}}$ and **92.33%** on $\text{TEST}_{\text{priv,mix}}$. These results are remarkably strong for the zero-shot setting, and even compare favorably with several methods developed under regular supervised or less constrained settings. This observation highlights the effectiveness of our threshold-calibrated boundary modeling for defect localization, especially in categories with fine-grained texture anomalies.

Overall, the results demonstrate that our method can produce accurate and reliable binary anomaly segmentation without using any target-domain training or validation data,

while maintaining consistent performance across different test conditions.

3.2. Comparison

Our method shows strong competitiveness against several conventional anomaly detection methods developed under regular setting. In particular, on $\text{TEST}_{\text{priv}}$, our method surpasses MSFlow by **20.84%** in SegF1, and on $\text{TEST}_{\text{priv,mix}}$, it exceeds RD++ by **24.75%**. These results are especially noteworthy because our model operates under the more challenging zero-shot protocol, without using any target-domain training or validation data.

This comparison demonstrates that our strong performance comes from two complementary components. First, adapting a vision foundation model with threshold-calibrated optimization enables more reliable industrial anomaly localization and binary segmentation by explicitly aligning training with the inference-time decision boundary. Second, the SAM-Finer post-processing further refines the coarse binary anomaly masks by sharpening object-aware boundaries, which improves the spatial quality of the final segmentation results. Together, these two components allow our method to produce more accurate and reliable binarized anomaly masks than directly applying conventional anomaly detection methods, even when those methods are developed under less constrained regular-setting assumptions.

4. Discussion

4.1. Challenges and Solutions

The main challenge in this track is to obtain reliable binary anomaly masks under the zero-shot setting. Since no target-domain data are available, the decision boundary cannot be tuned for each category. To address this issue, we propose BALM method, which aligns model training with a fixed low threshold. At inference time, we further adopt a slightly lower threshold to improve defect recall, and apply SAM-Finer to refine the coarse binary masks. This design enables stable and practical binary anomaly segmentation under the zero-shot protocol.

4.2. Model Robustness and Adaptability

Our method shows strong robustness under the zero-shot setting. It achieves **42.64%** SegF1 on $\text{TEST}_{\text{priv}}$ and **41.45%** on $\text{TEST}_{\text{priv,mix}}$, with only a small performance gap between the two test sets. This indicates good stability under more challenging conditions such as illumination changes. Moreover, our method even outperforms several conventional methods developed under regular-setting.

4.3. Model Efficiency

Our inference pipeline resizes each input image to 672×672 and performs prediction with only a simple rotation-based

Table 1. Comparison of segmentation F_1 scores (%) across different methods on binarized outputs for $\text{TEST}_{\text{priv}}$ / $\text{TEST}_{\text{priv,mix}}$ sets.

Object	Ours	PatchCore	RD	RD++	EfficientAD	MSFlow	SimpleNet	DSR
Can	0.1 / 0.1	0.3 / 0.1	0.1 / 0.1	0.1 / 0.1	0.8 / 0.1	5.0 / 0.1	0.6 / 0.1	0.4 / 0.1
Fabric	94.1 / 92.33	11.5 / 9.8	2.6 / 2.2	2.9 / 2.3	7.6 / 1.0	22.0 / 4.1	21.6 / 10.2	7.9 / 5.0
Fruit Jelly	30.94 / 30.74	8.7 / 8.2	22.5 / 22.7	26.9 / 26.7	20.8 / 18.2	47.6 / 38.1	25.1 / 23.0	17.9 / 17.2
Rice	57.78 / 56.19	3.8 / 4.2	7.0 / 3.9	9.5 / 2.9	15.0 / 0.5	19.1 / 1.8	11.6 / 1.0	1.5 / 1.4
Sheet Metal	67.9 / 67.18	1.8 / 1.1	41.3 / 39.2	40.9 / 37.7	9.3 / 3.8	13.0 / 7.6	14.6 / 2.8	13.9 / 14.4
Vial	19.01 / 18.47	2.3 / 2.2	28.0 / 28.3	28.2 / 22.8	30.5 / 26.5	23.3 / 6.2	31.9 / 17.5	28.2 / 27.9
Wall Plugs	8.72 / 6.89	0.0 / 0.0	1.9 / 0.8	1.3 / 0.9	4.4 / 0.3	0.1 / 0.2	1.0 / 0.3	0.4 / 0.4
Walnuts	62.59 / 59.72	1.2 / 1.3	41.2 / 36.7	44.1 / 40.5	34.6 / 13.3	44.5 / 14.3	35.2 / 14.3	17.0 / 9.6
Mean	42.64 / 41.45	3.7 / 3.4	18.1 / 16.7	19.2 / 16.7	15.4 / 8.0	21.8 / 9.0	17.7 / 8.7	10.9 / 9.5

test-time augmentation. Compared with methods that split high-resolution images into multiple local patches for inference, our method avoids patch cropping and result merging, while still achieving strong segmentation performance. This makes the overall pipeline more practical for real-world industrial applications, where both accuracy and efficiency are important.

4.4. Zero-Shot Setting

Our method strictly follows the industrial zero-shot setting. First, no MVTec AD 2 training, validation, or test images are used during model training. The model is trained only on auxiliary data and is not optimized using any target-domain samples. Second, during testing, the model parameters remain fixed, and no test-time adaptation, fine-tuning, or category-specific calibration is performed. The same inference procedure and fixed threshold are applied across all test categories, ensuring compliance with the zero-shot evaluation protocol.

4.5. Future Work

The current model is trained mainly on synthetic anomaly data. Although this strategy provides scalable supervision without using target-domain samples, synthetic defects may still differ from real industrial anomalies in appearance, scale, and spatial distribution. In future work, we plan to improve the training data by combining large-scale synthetic anomalies with real anomaly samples from auxiliary non-target domains. Training on a broader and more diverse anomaly corpus may further improve the model’s representation capability and enhance its generalization to challenging real-world industrial scenarios.

5. Conclusion

We presented BALM, a threshold-calibrated framework for zero-shot industrial anomaly detection and segmentation. Our method addresses a central challenge of the zero-shot setting: producing reliable binary anomaly masks without

any target-domain training or validation data. By aligning training with a fixed low decision boundary and refining coarse masks with SAM-Finer, BALM improves the quality of practical defect localization.

Experiments on MVTec AD 2 demonstrate the effectiveness of the proposed design. BALM achieves **42.64%** SegF1 on $\text{TEST}_{\text{priv}}$ and **41.45%** on $\text{TEST}_{\text{priv,mix}}$, with only a small performance gap between the two test sets, indicating strong robustness under more challenging mixed conditions. The method also compares favorably with several conventional methods developed under regular-setting assumptions, highlighting the strength of threshold-calibrated foundation-model adaptation for zero-shot anomaly segmentation.

Overall, these results show that explicitly modeling the decision boundary is important for real-world industrial anomaly detection, where target-domain calibration is often unavailable. BALM provides an effective and practical solution for zero-shot inspection, and suggests that adapting foundation models with boundary-aware objectives is a promising direction for advancing anomaly detection in data-scarce settings.

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